THE P-TORSION POINTS OF AN ABELIAN VARIETY IN CHARACTERISTIC P NUMBER AT MOST P^G

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Allowed standard prerequisites (editorial).

- Lemma lemma-group-scheme-addition-tangent-vectors (canonical tag 0BF5). cited at proof lines 2077-2079. Derivative of [p] is multiplication by p on tangent space, hence zero.
- Definition more-algebra-definition-p-basis (07P1) and lemma more-algebra-lemma-p-basis (07P2). cited at proof lines 2089-2091. The p-basis of k(A)/k.
- Lemma algebra-lemma-derivative-zero-pth-power (canonical tag 031W). cited at proof lines 2092-2094. Each [p]^*(t_i) is a pth power.
- Lemma lemma-degree-multiplication-by-d (canonical tag 0BFG). cited at proof lines 2110-2112. Degree of [p] is p^{2g}.

Source reference. Original lemma 0BF5: sources/stacks-chow/tags.txt, source lines 14373–14373. Cited at proof lines 2077-2079 to see the derivative of [p] is multiplication by p, hence zero in characteristic p; load-bearing.

Source reference. Original lemma 0BFH: sources/stacks-chow/tags.txt, source lines 14385–14385. Cited parenthetically at proof lines 2079-2080 as a comparison for the derivative computation; navigation for the parallel case.

Source reference. Original definition 07P1: sources/stacks-chow/tags.txt, source lines 9781–9781. Cited at proof lines 2089-2091 to set up t_1, ..., t_g; load-bearing for the p-basis construction.

Source reference. Original lemma 07P2: sources/stacks-chow/tags.txt, source lines 9782–9782. Cited at proof lines 2089-2091 alongside 07P1; load-bearing for the p-basis construction.

Source reference. Original lemma 031W: sources/stacks-chow/tags.txt, source lines 3999–3999. Cited at proof lines 2092-2094 to see [p]^*(t_i) has a pth root; load-bearing for the factorization.

Source reference. Original lemma 0BFG: sources/stacks-chow/tags.txt, source lines 14384–14384. Cited at proof lines 2110-2112 to know [p] has degree p^{2g}, whence the degree split with pi_2; load-bearing for the cardinality bound.

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Lemma 0.2. *Let k be a field of characteristic $p > 0$. Let A be an abelian variety of dimension g over k. The fibre of $[p] : A \to A$ over 0 has at most p^g distinct points.*

Proof. To prove this, we may and do replace k by the algebraic closure. By Lemma 0BF5 the derivative of $[p]$ is multiplication by p as a map $T_{A/k,e} \to T_{A/k,e}$ and hence is zero (compare with proof of Lemma 0BFH). Since $[p]$ commutes with translation we conclude that the derivative of $[p]$ is everywhere zero, i.e., that the induced map $[p]^*\Omega_{A/k} \to \Omega_{A/k}$ is zero. Looking at generic points, we find that the corresponding map $[p]^* : k(A) \to k(A)$ of function fields induces the zero map on $\Omega_{k(A)/k}$. Let $t_1, \ldots, t_g$ be a p-basis of $k(A)$ over k (More on Algebra, Definition 07P1 and Lemma 07P2). Then $[p]^*(t_i)$ has a pth root by Algebra, Lemma 031W. We conclude that $k(A)[x_1, \ldots, x_g]/(x_1^p - t_1, \ldots, x_g^p - t_g)$ is a subextension of $[p]^* : k(A) \to k(A)$. Thus we can find an affine open $U \subset A$ such that $t_i \in \mathcal{O}_A(U)$ and $x_i \in \mathcal{O}_A([p]^{-1}(U))$. We obtain a factorization

$$[p]^{-1}(U) \xrightarrow{\pi_1} \mathrm{Spec}(\mathcal{O}(U)[x_1, \ldots, x_g]/(x_1^p - t_1, \ldots, x_g^p - t_g)) \xrightarrow{\pi_2} U$$

of $[p]$ over U. After shrinking U we may assume that π_1 is finite locally free (for example by generic flatness – actually it is already finite locally free in our case). By Lemma 0BFG we see that $[p]$ has degree p^{2g}. Since π_2 has degree p^g we see that π_1 has degree p^g as well. The morphism π_2 is a universal homeomorphism hence the fibres are singletons. We conclude that the (set theoretic) fibres of $[p]^{-1}(U) \to U$ are the fibres of π_1. Hence they have at most p^g elements. Since $[p]$ is a homomorphism of group schemes over k, the fibre of $[p] : A(k) \to A(k)$ has the same cardinality for every $a \in A(k)$ and the proof is complete. □